Name:________________________________ Student ID:________________________

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question (1 point each, total of 35 points).

1) Which of the following statements regarding the demand for labour is <u>false</u>?
   A) If a firm is a monopolist in the product market, the firm's demand for labour is the same as the industry's demand for labour.
   B) If a firm is a monopolist in the product market, the firm's demand curve is more elastic than it would be if the firm were operating in a perfectly competitive product market.
   C) If a firm is a monopolist in the product market, it may or may not pay wages which are higher than the case in which the firm operates in a competitive output market.
   D) If a firm is a monopolist in the product market, it may still behave as a wage taker in the labour market.
   E) If the firm is a monopolist in the product market, the firm's demand curve is steeper than it would be if the firm were operating in a perfectly competitive product market.

2) If workers receive economic rents, then:
   A) They are receiving wages which are higher than the competitive level.
   B) They are receiving wages which are below the competitive level.
   C) If wages were to be increased, the employment level would rise.
   D) If wages were to be reduced, the firm would no longer be able to recruit labour.
   E) The product market could be perfectly competitive.

3) What is the primary difference between a monopsonist and a firm which operates in a perfectly competitive labour market?
   A) The monopsonist is likely to be more profitable.
   B) In order to recruit another worker, the monopsonist incurs a larger increase in labour costs.
   C) In order to recruit another worker, the monopsonist incurs a smaller increase in labour costs.
   D) The monopsonist hires such that the marginal revenue product of labour is equal to the marginal labour cost, but the non-monopsonist does not.
   E) The labour supply curve is irrelevant for the monopsonist.

4) The supply of labour to an industry is generally upward sloping because:
   A) Workers are more willing to work when there is the added incentives in the form of higher wages.
   B) It is more profitable for firms to hire more workers when labour is cheap.
   C) The substitution effect and the scale effect of a wage change work in the same direction.
   D) The wage elasticity of labour supply is elastic.
   E) There are many workers working for an industry.

5) Which of the following statements about a payroll tax is true?
   A) It generally reduces the quantity of workers.
   B) It is borne entirely by workers if the supply of labour is perfectly elastic.
   C) It reduces the quantity demanded of workers if the supply of labour is perfectly inelastic.
   D) It is borne entirely by workers if demand is perfectly inelastic.
   E) It is shared equally between workers and employers.

6) Which of the following statements is true? The burden of a payroll tax is paid:
   A) by workers if the supply of labour is perfectly elastic.
   B) by workers if the demand for labour is perfectly inelastic.
   C) by both workers and firms if the elasticities of demand and supply of labour are between zero and infinity.
   D) by firms if they are legally responsible for remitting the tax to the government.
   E) by workers if the taxes are withheld from their paychecks government.

7) What is the expression for the wage elasticity for a linear demand curve?
   A) W / N^d
   B) $\Delta W / \Delta N^d$
   C) $\Delta N^d / \Delta W$
   D) $(\Delta N^d / N^d) / (\Delta W / W)$
   E) $(\Delta W / W) / (\Delta N^d / N^d)$

8) Consider the diagram for a labour market in which the relevant job attribute is job safety (increasing from left to right). Why are the iso-profit curves concave, i.e., bowed in towards the origin?
   A) Because a firm's profit decreases if it provides a safer environment for the workers.
   B) Because paying higher wages and providing a safer environment for the workers are both costly for the employer.
   C) Because the marginal rate of transformation between wages and safety is increasing as a firm introduces stronger safety procedures.
   D) Because there is a negative trade-off between job safety and wages
   E) Because the marginal rate of transformation between wages and safety is decreasing as a firm introduces stronger safety procedures.

9) Which of the following is <u>not</u> a way in which high wages may induce higher productivity?
   A) Through improved morale
   B) Through greater intensity of work effort
   C) Through reductions in absenteeism
   D) Through higher levels of job safety and improved working conditions
   E) Through lower levels of turn over

10) The curvature of the iso-utility curve illustrates:
   A) a diminishing marginal rate of transformation between wages and job safety.
   B) a diminishing marginal rate of substitution between wages and job safety.
   C) the diminishing marginal rate of technical substitution between wages and job safety.
   D) the level of safety that can be exchanged for wages on the labour market.
   E) the law of diminishing marginal returns applied to wages.

11) Fill in the blanks in the sentence below. The more ________ the individual worker, the more he/she requires a _________ compensating wage differential in return for accepting a riskier work environment.
   A) risk-averse; larger
   B) risk-loving; larger
   C) risk-averse; smaller
   D) risk-neutral; smaller
   E) risk-neutral; larger

12) Fill in the blank in the following sentence. Higher iso-profit schedules would imply profits that are _________ the normal, competitive level.
   A) below
   B) higher than
   C) equal to
   D) greater than or equal to
   E) None of the answer choices are correct.

13) The slope of the wage-safety locus gives:
   A) The change in the wage premium that the labour market yields for differences in the salary of each worker.
   B) The change in the wage premium that the labour market yields for differences in the degree of risk of the job.
   C) The change in the wage premium that the labour market yields for differences in the utility levels of workers.
   D) The change in profitability that is associated with different combinations of wages and job safety.
   E) The change in the wage premium that the labour market yields for differences in the preference of the workers.

14) Which of the classical economists was the first to think and write about the topic of compensating wage differentials?
   A) David Ricardo
   B) Alfred Marshall
   C) John Stuart Mill
   D) John Maynard Keynes
   E) Adam Smith

15) Which of the following regarding attracting doctors to rural community is NOT correct?
   A) Most of the doctors prefer to work and live in the urban area.
   B) Doctors will not respond to salary premiums for practicing in these areas.
   C) Given the tight budget constraints, government cannot afford to pay the doctors a premium to attract them in the rural community.
   D) Rural communities do not always have fully staffed and equipped hospitals.
   E) The theory of compensating wages explains that the doctors will move to rural community as long as a premium is paid.

16) In the model of compensating differentials, the derivation of the market-wage safety locus is premised on which of the following conditions?
   A) perfect information and imperfectly competitive conditions
   B) Homogenous firms and homogeneous preferences of workers
   C) Risk neutral workers
   D) Perfect information and competitive conditions
   E) None of these above

17) Consider an equilibrium between a firm and a worker that consists of a relatively low wage and a relatively high degree of safety. How would one describe the parties?
   A) A relatively safe firm and a relatively risk-averse worker
   B) A relatively unsafe firm and a relatively risk-averse worker
   C) A relatively safe firm and a not so risk-averse worker?
   D) A relatively unsafe firm and a not so risk-averse worker?
   E) We cannot tell given this information

18) In the context of compensating differentials, what has the literature regarding UBER drivers found?
   A) That the drivers are underpaid based on their working conditions
   B) That the theory of compensating differentials does not application at all
   C) That the drivers' preferences are similar to that of cabdrivers engaged in standard jobs
   D) That they are willing to accept less favourable working conditions in exchange for higher wages
   E) That the drivers are willing to accept lower wages in exchange for flexibility

19) All of the following items are something that workers might receive or pay a compensating differential for except:
   A) The risk of injury
   B) A higher or lower level of utility
   C) Health benefits
   D) Family friendly work practices
   E) Employment security

20) According to the signaling theory of education,
   A) The level of education or training directly measures the productivity of the worker.
   B) The level of education or training of one worker affects the productivity of her coworkers.
   C) Education and training function as utility-generating goods from the worker's perspective.
   D) Education and training do not contribute directly to productivity, but act as a filter to screen candidates.
   E) The social rate of return to education is greater than the private rate of return.

21) The essence of human capital theory is that:
   A) Firms spend money on their capital stock and equipment in order for labour to become more productive.
   B) Investments are made in human resources so as to improve their productivity and, therefore, their earnings.
   C) All workers who want to increase their salary should obtain a university diploma.
   D) Workers invested in higher education in order to send a signal to potential employers regarding their underlying ability.
   E) Education and training do not contribute directly to productivity, but act as a filter to screen candidates.

22) The opportunity cost of obtaining human capital is:
   A) the job experience lost while attending school.
   B) the direct cost of books, supplies, and tuition fees.
   C) the income foregone while acquiring it.
   D) the estimate of the increase in salary that results from the acquisition of skills.
   E) Higher the greater the discount rate

23) Private costs and benefits are those that are:
   A) accrued or incurred by society.
   B) not entered into the calculation of the value of human capital.
   C) accrued or incurred by the parties investing in human capital.
   D) external to the decision of investing in human capital.
   E) of great concern to the government.

24) The shape of the "age-earnings" profiles reflect all of the following, except that:

A) earnings tend to increase with age but at a deceasing rate.
B) individuals generally continue to make human capital investments in the form of on-the-job training and work experience once they have entered the labour force.
C) the earnings of individuals with higher levels of education are generally higher than those with fewer years of education.
D) older, more experienced workers have higher wage profiles than younger workers.
E) None of these above, as they are all valid statements.

25) In the context of investments in human capital, perfect capital markets imply all of the following, except that:

A) An individual considering an investment in human capital does not have to be concerned with the timing of income.
B) An individual considering an investment in human capital does not have to be concerned with the timing of expenditures.
C) An individual can base his/her human capital decision on total lifetime income.
D) An individual faces no liquidity constraints as they can borrow against expected future income.
E) An individual makes an investment decision based on his/her current level of income.

26) The rule for optimal human capital investment is that:

A) the individual should increase years of education until the discounted present value of the benefits of an additional year of education equals the discounted present value of the additional costs.
B) the individual should increase years of education until the discounted present value of the benefits of an additional year of education is less than the discounted present value of the additional costs.
C) the individual should increase years of education until the discounted present value of the benefits of an additional year of education is greater than the discounted present value of the additional costs.
D) the individual should increase years of education until the discounted present value of the benefits of all of the years of education equals the discounted present value of all of the associated costs.
E) The individual should invest in training and skill development such that the potential employer is impressed with her credentials.

27) When some important variables that enter into economic decision making are not observable until after a decision has taken place, firms are said to operate in an environment of:

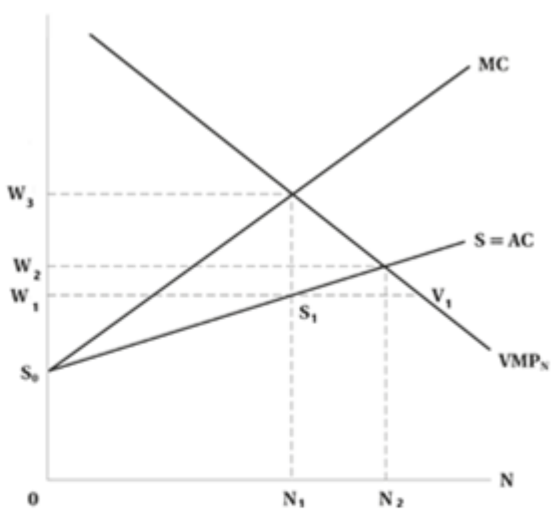
A) imperfect information.
B) market equilibrium.
C) imperfect competition in the output market.
D) imperfect competition in the input market.
E) perfect capital markets

Name:___________________________________ Student ID:__________________________

28) Fill in the blanks in the following sentence. Earnings tend to ________ with age, and thus with labour market experience, until age ________ years is reached, and then tend to ________.
   A) decrease; 40 to 50; increase
   B) decrease; 45 to 50; increase
   C) increase; 40 to 50; decline
   D) increase; 45 to 55; increase
   E) Level off; 40 to 50; decrease

29) In the context of discounted present value, under what conditions can the sum of the income earned in each period divided by the discounting factor be replaced by the payment divided by the rate of interest?
   A) When the stream of payments is collected over a long period of time
   B) When the interest rate is high
   C) When the payments are variable over time
   D) When the costs of obtaining the education are zero
   E) Under no circumstances is that approximation correct

Use the following diagram to answer questions 30-32:



30) VMP = value of marginal product, N = employment, and W = wage. The equilibrium wage and employment levels for the monopsonist are:
   A) W_3 and N_1
   B) W_1 and N_1
   C) W_2 and N_2
   D) W_1 and N_2
   E) W_2 and N_1

Name:__________________________________ Student ID:__________________________

31) VMP = value of marginal product, N = employment, and W = wage. The equilibrium wage and employment levels for the firm in a perfectly competitive labour market are:

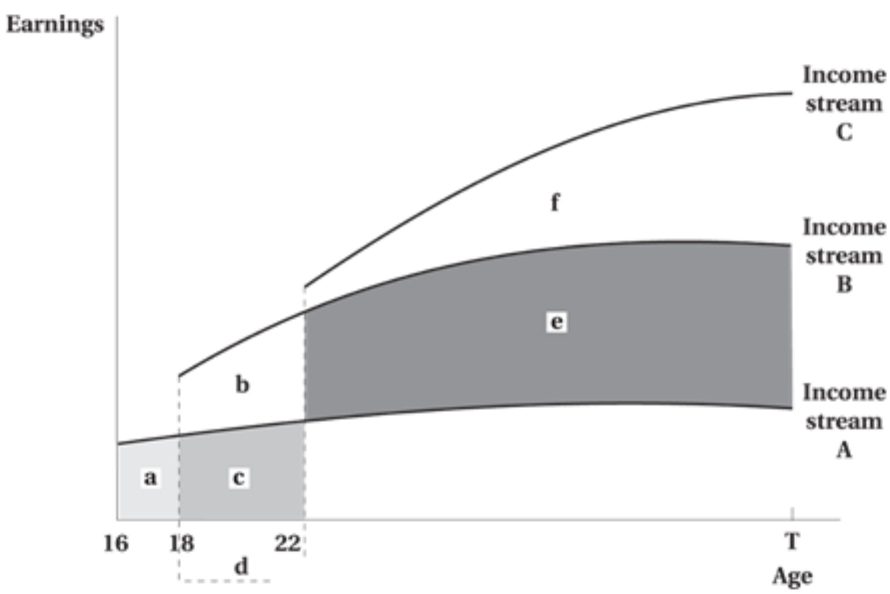
A) VMP_W and N_M.
B) W_M and N_M.
C) W_C and N_C.
D) W_M and N_C.
E) W_C and N_M.

32) VMP = value of marginal product, N = employment, and W = wage. The number of vacancies that exists for the monopsonist is:

A) Zero
B) Given by the distance W_CW_M
C) Given by the distance N_MN_C
D) Given by the distance V_MS_M
E) Given by distance S_MW_M

Use the following diagram to answer questions 33-35:

This diagram shows the age-earnings profiles of three different individuals who opt to choose three options: not graduating from high school, graduating from high school, and graduating from a university.



33) The area labeled d) refers to:
   A) The explicit, out-of-pocket costs incurred
   B) The opportunity costs incurred by someone who opted for income stream B (a high school graduate)
   C) The earnings premium earned by someone who opted for income stream B (a high school graduate) relative to a worker who did not obtain a high school diploma.
   D) The earnings premium earned by someone who opted for income stream C (a university graduate) relative to a worker who graduated from high school.
   E) The total cost of attending university

34) The area labeled e) refers to:
   A) The explicit, out-of-pocket costs incurred
   B) The opportunity costs incurred by someone who opted for income stream B (a high school graduate)
   C) The earnings premium earned by someone who opted for income stream B (a high school graduate) relative to a worker who did not obtain a high school diploma.
   D) The earnings premium earned by someone who opted for income stream C (a university graduate) relative to a worker who graduated from high school.
   E) The total cost of attending university

35) The sum of the areas labeled b), c), and d) refers to:
   A) The explicit, out-of-pocket costs incurred
   B) The opportunity costs incurred by someone who opted for income stream B (a high school graduate)
   C) The earnings premium earned by someone who opted for income stream B (a high school graduate) relative to a worker who did not obtain a high school diploma.
   D) The earnings premium earned by someone who opted for income stream C (a university graduate) relative to a worker who graduated from high school.
   E) The total cost of attending university